Exam Problem 2 • Multi-Year Horizon

INFOPYMES – Full Cycle Financial Statements, 2016–2017 & ROA/ROE

Original statement · T3_Additional_EXERCISES.pdf, p. 3

Three entrepreneurs founded **INFOPYMES** on 1 January 2016 with 100,000 € equity capital. A bank granted a 5-year loan of 100,000 € at 10% annual interest, returning 20,000 € principal each 31 December plus due interest. Computer equipment was bought on 01/01/2016 for 100,000 € (paid 01/04/2016; 5-year life, linear, 0 residual value). Online revenues: 300,000 €. Premises rent: 1,000 €/month. Permanent staff: 5,000 €/month. Utilities: 500 €/month. Variable consulting staff: 174,000 €. Results declared at 23:59 on 31 December; corporate tax 30% paid immediately; dividends 40% paid to partners.

1. Construct 2016 Income Statement.

2. Construct 31/12/2016 Balance Sheet at 24:00.

3. Construct 31/12/2017 Balance Sheet at 24:00 (assume no new investments, cash and retained earnings unchanged).

4. Determine ROA and ROE in 2017.

EXAM DAY STRATEGY: UNDERSTANDING "AT 24:00 AFTER PAYMENTS"

Because the prompt specifies that taxes, loan repayments, and dividends are paid *immediately before 24:00* on 31 December:

- No corporate taxes payable appear on the 2016 balance sheet.
- No dividends payable appear on the 2016 balance sheet.
- The bank loan principal drops immediately from €100,000 to €80,000.
- In 2017, the second repayment drops the loan principal from €80,000 to €60,000, while accumulated depreciation reaches €40,000.

Calculations & Financial Statements

1. **2016 Income Statement:**
   - Operating Expenses: Rent €12,000 + Permanent Staff €60,000 + Utilities €6,000 + Variable Wages €174,000 + Depreciation €20,000 (100k/5) = **€272,000**.
   - $\text{EBIT} = 300,000 - 272,000 = \mathbf{28,000}$ €.
   - Financial Interest (10% of €100,000): $\mathbf{10,000}$ €.
   - $\text{EBT} = 28,000 - 10,000 = \mathbf{18,000}$ €.
   - Tax (30%): $0.30 \times 18,000 = \mathbf{5,400}$ €.

- Net Income $= \mathbf{12,600}$ € $\implies$ Dividends $(40\%) = \mathbf{5,040}$ €, Retained Earnings $(60\%) = \mathbf{7,560}$ €
.

2. **Ending Cash at 31/12/2016:**

$$\text{Inflows } (100\text{k} + 100\text{k} + 300\text{k}) - \text{Outflows } (100\text{k} + 12\text{k} + 60\text{k} + 6\text{k} + 174\text{k} + 20\text{k} + 10\text{k} + 5.4\text{k} + 5.04\text{k}) = \mathbf{107,560} \text{ €}$$

(In the outflows, the second 20k is the 31 December loan principal repayment, not depreciation.)

BALANCE SHEET COMPARISON	At 31/12/2016 (€)	At 31/12/2017 (€)	Account Interpretation
• Computer Equipment (Cost)	100,000	100,000	Fixed capital asset
• (-) Accumulated Depreciation	(20,000)	(40,000)	1 year vs 2 years linear wear
TOTAL NON-CURRENT ASSETS	**80,000**	**60,000**	**Net Book Value**
• Cash & Liquid Treasury	107,560	107,560	Stated unchanged by prompt
TOTAL ASSETS	**187,560**	**167,560**	**Balanced Asset Base**
• Subscribed Share Capital	100,000	100,000	Partners' equity
• Retained Earnings	7,560	7,560	Stated unchanged by prompt
TOTAL OWNERS' EQUITY	**107,560**	**107,560**	**Net Worth**
• Long-term Bank Debt	60,000	40,000	Remaining balance > 1 year
• Short-term Bank Debt	20,000	20,000	Principal due next 31 Dec
TOTAL LIABILITIES & EQUITY	**187,560**	**167,560**	**Perfect Match!**

Determination of ROA and ROE in 2017

$$\text{ROA}_{\mathbf{2017}} = \frac{\text{EBIT}_{2017}}{\text{Total Assets}_{2017}} = \frac{28,000 \text{ €}}{167,560 \text{ €}} \approx \mathbf{16.71\%}$$

$$\text{ROE}_{\mathbf{2017}} = \frac{\text{Net Income}_{2017}}{\text{Total Equity}_{2017}} = \frac{14,000 \text{ €}}{107,560 \text{ €}} \approx \mathbf{13.02\%}$$

(Where $\text{Net Income}_{2017} = (28,000 - 10\% \times 80,000) \times (1 - 0.30) = 20,000 \times 0.70 = 14,000$ €*. This assumes that 2017 revenues and operating costs equal those of 2016 and that interest is* 10% *of the* $80,000$ *outstanding at the start of 2017. The statement also fixes retained earnings and cash at their 2016 values, so the 2017 net income of* $14,000$ *is not reflected as an increase in the balance sheet reserves.)*

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. 2016 is the first year; revenue is fully collected and all expenses are paid in the year. Interest is 10% of the balance at the start of the year (100,000 in 2016, 80,000 in 2017); 20,000 of principal is repaid each 31 December; tax (30% of EBT) and the 40% dividend are paid at 23:59 on 31 December, so

no tax or dividend liability remains at 24:00. For 2017 the statement imposes unchanged cash and retained earnings and no new investment; operating results are assumed equal to 2016.

Phase 2: Definitions. $A = L + E$; $\text{ROA} = \text{EBIT}/\text{Total assets}$; $\text{ROE} = \text{Net income}/\text{Equity}$; equity = capital + retained earnings, with retained earnings = net income - dividends.

Phase 3: Journal entries 2016 (€). Each entry was balanced in Python (debits = credits).

#	Event	Debit	Credit
1	1 Jan: partners' capital	Cash 100,000	Capital 100,000
2	1 Jan: bank loan 5 years at 10%	Cash 100,000	Bank loan 100,000
3	1 Jan: computer equipment bought on credit (payable 1 Apr)	Equipment 100,000	Equipment payable 100,000
4	1 Apr: equipment supplier paid	Equipment payable 100,000	Cash 100,000
5	2016 consulting revenue, fully collected	Cash 300,000	Revenue 300,000
6	Rent 12 x 1,000	Rent 12,000	Cash 12,000
7	Permanent staff and Social Security 12 x 5,000	Staff 60,000	Cash 60,000
8	Electricity, phone 12 x 500	Utilities 6,000	Cash 6,000
9	Variable wages of consulting staff	Variable wages 174,000	Cash 174,000
10	Depreciation 100,000 / 5	Depreciation 20,000	Accum. depreciation 20,000
11	31 Dec: interest 10% x 100,000 paid	Interest 10,000	Cash 10,000
12	31 Dec: loan principal repayment	Bank loan 20,000	Cash 20,000
13	31 Dec: corporate tax 30% x 18,000 paid	Tax 5,400	Cash 5,400
14	31 Dec: dividends 40% x 12,600 paid	Dividends 5,040	Cash 5,040

Closing ledger account (24:00, 31/12/2016)	Balance (€)
Cash	107,560 Dr
Equipment	100,000 Dr
Accum. depreciation	20,000 Cr
Bank loan	80,000 Cr
Capital	100,000 Cr

Step-by-step lines: $\text{EBIT} = 300{,}000 - (12{,}000 + 60{,}000 + 6{,}000 + 174{,}000 + 20{,}000) = 28{,}000$; $\text{EBT} = 28{,}000 - 10{,}000 = 18{,}000$; $\text{tax} = 0.30 \times 18{,}000 = 5{,}400$; net income $= 12{,}600$; dividends $= 0.40 \times 12{,}600 = 5{,}040$; retained earnings $= 7{,}560$ (closed from income and dividends, so not a ledger line above). Balance 2016: $80{,}000 + 107{,}560 = 187{,}560 = 80{,}000 + 100{,}000 + 7{,}560$.

2017 bridge. Interest $= 0.10 \times 80{,}000 = 8{,}000$; $\text{EBT} = 20{,}000$; tax $= 6{,}000$; net income $= 14{,}000$. Cash after interest and tax $= 28{,}000 + 20{,}000 - 8{,}000 - 6{,}000 = 34{,}000$; minus principal 20,000 gives 14,000, which equals the dividend if the whole net income is distributed. Hence $\Delta\text{Cash} = 14{,}000 - 14{,}000 = 0$ and

$\Delta \text{RE} = 14{,}000 - 14{,}000 = 0$, consistent with the statement. Ratios: $\text{ROA} = 28{,}000/167{,}560 = 16.71\%$; $\text{ROE} = 14{,}000/107{,}560 = 13.02\%$.

Phase 4: Units and sanity checks. Currency: euro (€); ratios are dimensionless. $A = L + E$: $187{,}560 = 80{,}000 + 107{,}560$ (2016) and $167{,}560 = 60{,}000 + 107{,}560$ (2017). Cash identity 2016: $500{,}000 - 392{,}440 = 107{,}560$. ROE is below ROA because interest and tax are deducted from its numerator, even though the equity base is smaller than total assets. No official numerical key is available in `sources/` (only the statement PDF).

Study notes written from the official course material (T3_Additional_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-extra-exam-problems.html